Resolving a category error: DHL[3, 2] yields $H_1 \leq 6$, not $H_1 = 2$

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Let $H_1 = \liminf_{n\to\infty}(p_{n+1}-p_n)$. Write DHL$[k, j]$ for the assertion that every admissible k-tuple of linear forms has infinitely many translates containing at least j primes.

Disclaimer: This note does not prove the hypothesis DHL[3, 2], nor does it prove $H_1 = 2$. It maps the strict logical implications of those statements.

The implication DHL$[3, 2] \Rightarrow H_1 \leq 6$ is elementary and unconditional. If at least two elements of an admissible 3-tuple of diameter 6 are prime for infinitely many translates, then infinitely many prime pairs differ by at most 6, and therefore $H_1 \leq 6$.

This is not the Twin Prime Conjecture. After the single odd prime gap $3 - 2 = 1$, every prime gap is even. Therefore, the inequality $H_1 \leq 3$ is logically equivalent to $H_1 = 2$. The statement DHL[3, 2] produces prime differences in the set $\{2, 4, 6\}$; it does not force the consecutive-gap limit inferior to equal 2. Conflating DHL[3, 2] with $H_1 \leq 3$ is a category error.

The implication DHL$[3, 2] \Rightarrow H_1 \leq 6$ does not require the generalized Elliott–Halberstam conjecture (GEH). However, the hypothesis DHL[3, 2] itself is presently known only under GEH, which is how the conditional bound $H_1 \leq 6$ is obtained. By the widely accepted parity barrier (going back to Selberg), standard square-of-divisor-sums sieve architectures are not expected to go below 6, even under GEH; in this note that barrier is used only as an explicit hypothesis (Theorem 4), not proved.

The long-standing unconditional bound remains Polymath8b (2014), which established $H_1 \leq 246$ using the Maynard–Tao sieve on an admissible 50-tuple. A claimed improvement to $H_1 \leq 240$, attributed to a 2026 preprint by Stadlmann, is **[citation needed: unverified]** and is not relied upon here.

The two statements that can be written without ambiguity are therefore:

- DHL[3, 2] (presently conditional on GEH) implies $H_1 \leq 6$.
- $H_1 = 2$ (the Twin Prime Conjecture).

Theorem 1: The trivial lower bound $H_1 \geq 2$

Theorem 1. *$H_1 \geq 2$. Consequently, any claim that $H_1 \leq 3$ is exactly equivalent to $H_1 = 2$.*

Proof. The sequence of primes begins $p_1 = 2, p_2 = 3, p_3 = 5, p_4 = 7, \ldots$. For the first pair, the consecutive gap is $p_2 - p_1 = 3 - 2 = 1$. Because 2 is the only even prime, every prime p_n for $n \geq 2$ is odd. The difference between any two odd integers is necessarily an even integer.

Since consecutive primes are distinct, $p_{n+1} - p_n > 0$. The smallest positive even integer is 2. Therefore, for all $n \geq 2$, $p_{n+1} - p_n \geq 2$. Taking the limit inferior of this sequence yields $H_1 \geq 2$.

Because the only integers in the interval $[2, 3]$ are 2 and 3, and 3 is impossible for $n \geq 2$, $H_1 \leq 3$ collapses to $H_1 = 2$. □

Theorem 2: Proof of the implication $\mathrm{DHL}[3, 2] \Rightarrow H_1 \leq 6$

Theorem 2. *Assume* $\mathrm{DHL}[3, 2]$. *Then* $H_1 \leq 6$.

Proof. By definition, $\mathrm{DHL}[3, 2]$ asserts that every admissible 3-tuple has infinitely many translates containing at least two primes. Two equally narrow admissible 3-tuples exist, and both act as equal witnesses to this bound: $\mathcal{H}_1 = \{0, 2, 6\}$ and $\mathcal{H}_2 = \{0, 4, 6\}$.

We first verify from first principles that $\mathcal{H}_2 = \{0, 4, 6\}$ is strictly admissible by confirming it does not occupy all residue classes for any prime p:

- For $p = 2$: The elements modulo 2 are $\{0, 0, 0\}$. The residue class $1 \pmod 2$ is omitted.
- For $p = 3$: The elements modulo 3 are $\{0, 1, 0\}$. The residue class $2 \pmod 3$ is omitted.
- For $p \geq 5$: The tuple contains exactly 3 elements. Since $3 < p$, the Pigeonhole Principle dictates these elements can occupy at most 3 distinct residue classes. At least $p - 3 \geq 2$ residue classes modulo p must remain empty.

Because $\mathcal{H}_2$ is admissible and has a diameter of $6 - 0 = 6$, the hypothesis $\mathrm{DHL}[3, 2]$ guarantees that there exist infinitely many integers n such that the translated tuple $(n, n + 4, n + 6)$ contains at least two primes.

For each such n, the two primes must manifest as one of the following pairs:

- $(n, n + 4)$, yielding a prime difference of 4;
- $(n + 4, n + 6)$, yielding a prime difference of 2;
- $(n, n + 6)$, yielding a prime difference of 6.

In all three cases, we obtain two primes p and q $(p < q)$ satisfying $q - p \leq 6$. Write $p = p_k$. The next prime after p is at most q, since q itself is a prime larger than p; hence $p_{k+1} - p_k \leq q - p \leq 6$. Moreover $p \geq n$, so as n grows the index k is unbounded.

Because this translates infinitely often across unbounded n, the sequence of such index pairs $(k, k + 1)$ is unbounded. The occurrences of gaps $p_{k+1} - p_k \leq 6$ are not confined to a finite initial segment of the primes. Therefore, by the formal definition of the limit inferior,

$$H_1 = \liminf_{k \to \infty} (p_{k+1} - p_k) \leq 6.$$

This completes the proof. □

Theorem 3: $\mathrm{DHL}[2, 2]$ does not follow from the combinatorial form of $\mathrm{DHL}[3, 2]$

For a predicate P on $\mathbb{N}$, write $\mathrm{DHL}_P[k, j]$ for the statement obtained from $\mathrm{DHL}[k, j]$ by replacing "prime" with "satisfies P".

Theorem 3. *Let* $P(n)$ *hold iff* $n \bmod 4 \in \{0, 1\}$. *Then* $\mathrm{DHL}_P[3, 2]$ *holds (indeed for every 3-element set, in particular* $\{0, 2, 6\}$ *and* $\{0, 4, 6\}$*), while* $P(n) \wedge P(n + 2)$ *holds for no* n*; hence* $\mathrm{DHL}_P[2, 2]$ *fails, because* $\{0, 2\}$ *is admissible.*

Proof. P never contains two integers differing by 2, since such integers have residues differing by 2 (mod 4). For any $d \not\equiv 2 \pmod 4$, P contains infinitely many pairs $(m, m+d)$: take $m \equiv 0 \pmod 4$ if $d \equiv 0, 1$ and $m \equiv 1$ if $d \equiv 3$. Among three distinct integers $a < b < c$, the differences $b - a$ and $c - b$ cannot both be $\equiv 2 \pmod 4$ without $c - a \equiv 0$. So some pair in the set has a difference $\not\equiv 2$ (mod 4), and that pair is hit infinitely often. □

Scope. This shows only that no proof of DHL$[3, 2] \Rightarrow$ DHL$[2, 2]$ can use the combinatorial form of DHL$[3, 2]$ alone; any such proof must use further properties of the primes. It is *not* an independence result for the actual primes. DHL$[3, 2]$ is fully compatible with $H_1 = 2$.

Theorem 4: The parity hypothesis and the minimal diameter 6

Parity hypothesis (assumed, not proved). Following the parity discussion in the Polymath8b project, we *assume* the following of a given sieve method: whenever it establishes DHL$[k, j]$, then $j \leq \lceil k/2 \rceil$. We use this only as an explicit hypothesis about square-of-divisor-sum sieve architectures, not as a theorem.

Theorem 4. *1. Under the parity hypothesis, a sieve establishing* DHL$[k, 2]$ *must have* $k \geq 3$.

2. (Unconditional.) Every admissible 3-element set $\mathcal{H} \subset \mathbb{N}$ *has diameter at least* 6*, and* $\{0, 2, 6\}$*,* $\{0, 4, 6\}$ *attain it. So the minimal diameter of an admissible 3-tuple is exactly* 6*.*

Proof. (1) $2 \leq \lceil k/2 \rceil$ forces $k \geq 3$. (2) Admissibility at 2 forces all three elements to share a parity. If the diameter were at most 5, the elements would be $m, m + 2, m + 4$, which occupy all residues mod 3, contradicting admissibility at 3. □

Under the parity hypothesis, sieve methods of this type therefore cannot get below $H_1 \leq 6$ via DHL$[k, 2]$.

The Frontier: Deploying Parity-Detecting Techniques

If a future manuscript claims to prove the Twin Prime Conjecture ($H_1 = 2$), its validity can be immediately triaged by locating its approach to the parity obstruction. A standard GPY/Maynard–Tao square-of-divisor-sums argument cannot achieve it alone.

A genuine proof needs an ingredient that explicitly distinguishes integers with $\Omega(n) = 1$ from those with $\Omega(n) = 2$. In other contexts of analytic number theory, mathematicians have successfully deployed precise parity-detecting capabilities by decomposing the von Mangoldt function via combinatorial identities and estimating the resulting multi-linear forms.

Notable examples of arguments that overcome or sidestep parity-type limitations include:

- **Friedlander and Iwaniec (1997):** Proved the existence of infinitely many primes of the form x^2+y^4 by successfully bounding Type II (bilinear) sums to bypass the parity obstruction in two-dimensional geometries.
- **Maynard (2019):** Proved there are infinitely many primes with missing digits using Fourier-analytic control of Type I and Type II sums over restricted digit sets. This is a related use of bilinear-sum technology rather than a direct parity-breaking result of the Friedlander–Iwaniec kind.

To progress from a bounded gap of 6 to a bounded gap of 2, a proof must exhibit this shape: it must successfully inject parity-breaking bilinear forms (Type II sums) directly into the linear geometry of bounded prime gaps. Until a technique exhibiting this precise capability is successfully deployed, sieving on $k = 3$ forms remains the minimum viable dimension, and $H_1 \leq 6$ remains the floor for the GPY/Maynard–Tao sieve architecture under the parity hypothesis.

Formal verification in Lean 4

All theorems above are machine-checked in Lean 4 + Mathlib (no `sorry`, no axioms beyond Lean's standard `propext`, `Classical.choice`, `Quot.sound`) in `https://github.com/BenFrohman/DHL-3-2-implies-liminf-prime-gap-le-6`. Here $H_1 \leq c$ is encoded as "$p_{n+1} - p_n \leq c$ for infinitely many n".

- Theorem 1: `DHL32.two_le_gap`, `DHL32.not_H1Le_one`, `DHL32.H1Le_three_iff_two`, `DHL32.H1Le_three_i`
- Admissibility: `DHL32.admissible_046`, `DHL32.admissible_026`, `DHL32.admissible_02`.
- Theorem 2: `DHL32.dhl32_H1Le_six`.
- Theorem 3: `DHL32.dhl32_not_implies_dhl22_combinatorially`, `DHL32.Ptoy_DHL32`.
- Theorem 4: `DHL32.ParityBarrier` (hypothesis structure), `DHL32.parity_two_needs_three`, `DHL32.admissible_three_diam_ge_six`, `DHL32.min_diam_admissible_three`.

Literature statements (Polymath8b's 246, the unverified Stadlmann claim, GEH conditionality of DHL$[3, 2]$) are not formalized.